

Final Forecast Model Report

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INF 499

Introduction and Objective

The goal of this project was to generate probabilistic forecasts for weekly influenza-like illness percentage, or ILI%, using the ATSF Forecast Hub dataset. ILI forecasting is important because flu activity affects healthcare systems each year, and short-term forecasts can help support planning and resource allocation. The task is challenging because ILI data show strong annual seasonality, but seasonal peaks vary in timing and magnitude across years. My original proposal identified this combination of clear seasonality and high variability as a main challenge of the forecasting problem.

Figure 1 shows the weekly ILI time series used for modeling. The repeated peaks show the annual flu cycle, while the changing peak heights show why forecast uncertainty is important. My objective was to implement a valid, reproducible forecasting model based on methods covered in class, evaluate its validation performance, and submit a complete forecast in the required hub format. My final submitted model was named **adk-stletsbc** for hub formatting purposes. During validation, this same model was tested as **adk-stlets-bc015**.

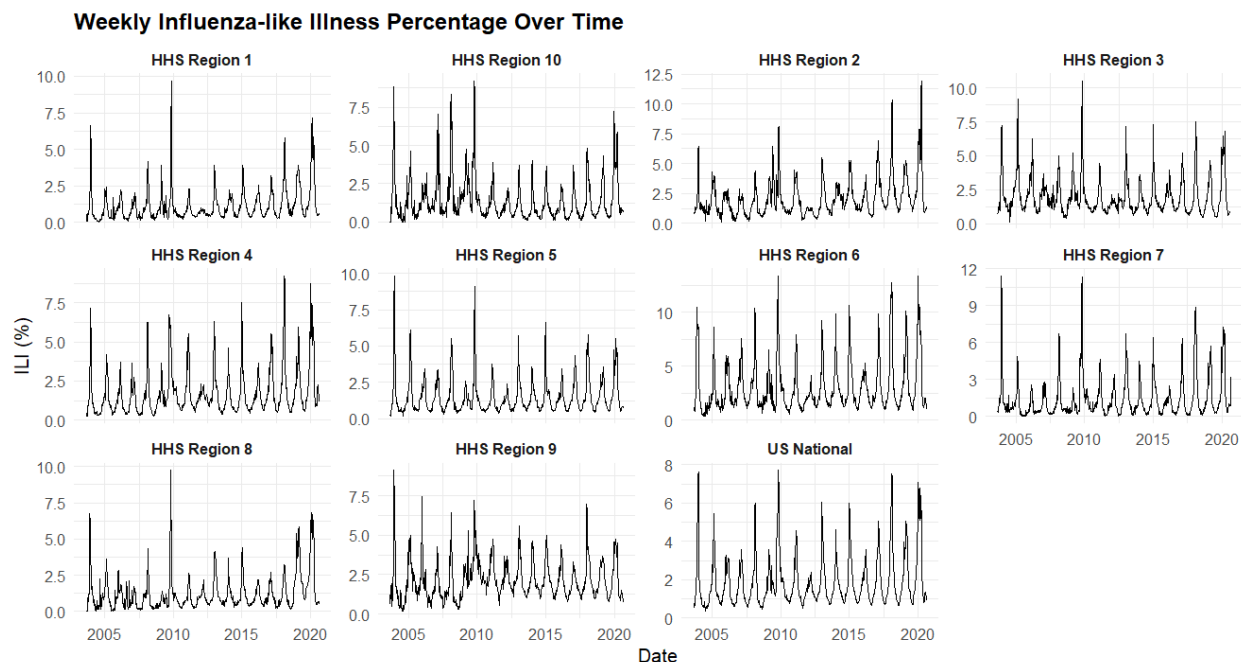


Figure 1. Weekly influenza-like illness percentage over time. The time series shows strong recurring seasonal patterns, with flu activity typically increasing during seasonal peaks. However, the magnitude of these peaks varies across years, which increases uncertainty and motivates the use of a seasonal forecasting model.

Methodology

My final model used **STL decomposition** followed by **ETS forecasting**. STL separates the ILI time series into trend, seasonal, and remainder components, which is useful because ILI has a strong annual seasonal pattern. After decomposition, an ETS model was fit to the seasonally adjusted series, and the forecast was recombined with the seasonal component.

The seasonal period was set to **52 weeks** to represent the annual flu cycle. I also applied a **Box-Cox transformation** before fitting the model to help stabilize variation between low-activity weeks and peak flu periods. I tested several related versions of the model, including no transformation, log, square-root, and Box-Cox transformations. The final submitted model used **Box-Cox $\lambda = 0.15$** because it produced the best validation WIS.

Forecasts were generated retrospectively for each required origin date. For each date, the model was trained only on observations available up to that point, then used to generate 1–4 week ahead forecasts. Simulated forecast paths were converted into the required quantiles and saved in the hub format with the correct locations, horizons, target end dates, output types, and quantile IDs.

Results

Performance was evaluated during the validation period using **weighted interval score** (WIS) along with 50% and 90% prediction interval coverage. WIS was the main selection metric because it evaluates both forecast accuracy and uncertainty, with lower values indicating better probabilistic forecasts.

The final model, **adk-stletsbc**, was evaluated during validation as **adk-stlets-bc015** and achieved the lowest WIS of the models tested: **0.2987**. Its 50% interval coverage was **0.5841**, and its 90% interval coverage was **0.9219**.

Table 1. Potential models WIS and 50% and 90% coverage performance.

Model	Validation WIS	50% Coverage	90% Coverage
adk-stlets-bc015	0.2987	0.5841	0.9219
adk-stlets-bc020	0.2989	0.5742	0.9163
adk-stlets-log	0.3002	0.5307	0.8720
adk-stlets-sqrt	0.3135	0.5522	0.8716
adk-stlets	0.3501	0.6152	0.9003
adk-rwdriftbs	0.3578	0.3736	0.8664

As shown in Table 1, the transformed STL + ETS models outperformed the untransformed STL + ETS model and the random walk with drift baseline. The Box-cox $\lambda = 0.15$

and $\lambda = 0.20$ models were very close, but $\lambda = 0.15$ had the best WIS overall. The interval coverage suggests that the final model was slightly conservative, but still reasonably calibrated.

Discussion and Conclusions

Overall, **adk-stletsbc** performed well for the goals of this project. The STL + ETS structure was appropriate because ILI has a recurring annual pattern, and decomposition allowed the model to separate that seasonal pattern from shorter-term variation. The Box-Cox transformation was the most useful improvement, reducing WIS from **0.3501** for the untransformed STL + ETS model to **0.2987** for the final model.

The main limitation is that the model only used historical ILI% values. It did not include external predictors such as weather, school calendars, vaccination patterns, mobility, or behavior. It also assumes that seasonal structure is fairly stable, which may not hold during unusual flu seasons or sudden outbreaks.

Future improvements could include testing SARIMA, regression with seasonal terms, or ensembles. However, for this INF 499 submission, **adk-stletsbc** was an appropriate and reproducible course-based model. It passed the required hub validation checks and achieved the best validation WIS among the models I tested.

References

Hyndman, R. J. & Athanasopoulos, G. (2021). *Forecasting Principles and Practice* (3rd Edition). OTexts.

Centers for Disease Control and Prevention. (2026). *FluSight*. U.S. Department of Health and Human Services. <https://www.cdc.gov/flu-forecasting/index.html>